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Mr. Learner

AUTHBLOCK

An Enterprise-Grade Decentralized Application for

Academic Credential Verification Using Blockchain

A Mini-Project Report Submitted

For

Partial Fulfilment of the Requirements of the

Degree of Bachelor of Engineering

In

COMPUTER ENGINEERING

(Semester VI)

By

Jason Gonsalves (10246)

Ashley Almeida (10227)

Under the guidance of

Prof. Prajakta Dhamanskar

2025 2026

This work is dedicated to our families.

We are very thankful for their motivation and support.

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CERTIFICATE

This is to certify that the mini-project entitled "AUTHBLOCK" is a bonafide work of Jason Gonsalves (10246), Ashley Almeida (10227) submitted to the University of Mumbai in partial fulfillment of the requirement for the award of the degree of Bachelor of Engineering in Computer Engineering (Semester-VI).

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Approval Sheet

Mini Project Report Approval for T.E.

This mini-project report entitled "AUTHBLOCK" submitted by Jason Gonsalves (10246), Ashley Almeida (10227) is approved for the degree of Bachelor of Engineering in Computer Engineering (Semester-VI).

Examiner 1: _____

Examiner 2: _____

Date:

Place: Mumbai

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Declaration

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources.

We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission.

We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

Jason Gonsalves (10246) _____

Ashley Almeida (10227) _____

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Abstract

Academic credential fraud is a growing global concern, undermining the integrity of educational institutions and professional hiring processes.

Traditional certificate verification methods rely on centralized systems prone to tampering, data breaches, and inefficient manual processes.

This mini-project, Authblock, presents an enterprise-grade decentralized application (dApp) that leverages the Ethereum blockchain to provide a tamper-proof, end-to-end solution for generating, storing, and verifying academic marksheets and certificates.

Authblock implements a multi-layered architecture integrating Next.js 14 on the frontend, serverless PostgreSQL (NeonDB) for off-chain data persistence, Amazon Web Services (AWS) for secure cloud storage and event-driven communications, and a custom Solidity smart contract deployed on the Ethereum Sepolia Testnet for immutable hash anchoring.

The workflow begins with an authorized administrator uploading student result data via CSV or manual entry. The system dynamically generates PDF marksheets using jsPDF and html2canvas, uploads them to Amazon S3, and computes a SHA-256 hash of the document content. This hash is written to the Ethereum blockchain, creating an immutable record of the document's provenance.

Multi-Factor Authentication via Firebase Auth ensures that only authorized personnel can issue credentials. Students receive automated email notifications through an event-driven pipeline consisting of Amazon SNS, SQS, and Lambda, which triggers Amazon SES for email delivery.

Verification is accessible to any third party by comparing a document's computed hash against the on-chain record, instantly revealing any tampering or forgery.

The system also incorporates optional OCR-based verification using OCR.space and Google Cloud Vision APIs for scanned document matching.

Security is reinforced through IAM Principle of Least Privilege policies, strict environment key management, and continuous monitoring via AWS CloudWatch and CloudTrail.

The result is a scalable, trustless, and transparent credential ecosystem that eliminates academic fraud, reduces verification turnaround time, and builds institutional credibility.

Keywords: Blockchain, Ethereum, Academic Credentials, Smart Contracts, Decentralized Application, AWS, Document Verification, SHA-256, NFT, Next.js, Firebase Authentication, Amazon S3, Credential Fraud Prevention

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We also thank all non-teaching staff for their valuable support in completing our project.

We extend our gratitude to the open-source community behind Next.js, Ethereum, and AWS documentation which greatly aided our learning and development.

Date:

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Chapter 1

INTRODUCTION

In the contemporary digital era, academic institutions issue thousands of certificates, marksheets, and degrees every year.

These documents are fundamental to a student's career trajectory, yet they remain susceptible to forgery, alteration, and misuse.

A 2022 report by the Higher Education Degree Datacheck (HEDD) found that approximately one in ten candidates misrepresent their academic qualifications during job applications.

This alarming statistic highlights the inadequacy of conventional centralized verification systems.

Centralized systems store credential data in single-point databases controlled by individual institutions.

These are vulnerable to insider threats, data breaches, and unauthorized modifications.

Furthermore, verifying credentials through these systems typically requires contacting the issuing institution directly - a process that is time-consuming, geographically constrained, and often prone to human error.

Third parties such as employers or immigration authorities face significant friction when verifying academic claims.

Blockchain technology offers a transformative alternative.

A blockchain is a distributed, append-only ledger where data once written cannot be altered without consensus from the network.

By anchoring cryptographic hashes of academic documents onto a public blockchain, institutions can make credential verification instantaneous, borderless, and trustless.

Any third party can independently verify a document's authenticity simply by recomputing its hash and comparing it against the on-chain record - no institutional contact required.

Authblock is an enterprise-grade decentralized application (dApp) designed to address these challenges comprehensively.

It integrates blockchain technology with modern cloud infrastructure to provide a complete pipeline for credential issuance, storage, and verification.

The system is built on Next.js 14 with a React 18 frontend, a serverless NeonDB PostgreSQL database for off-chain metadata, AWS cloud services (S3, SNS, SQS, Lambda, SES) for storage and notifications, and a Solidity smart contract deployed on the Ethereum Sepolia Testnet for immutable hash anchoring.

The system supports two primary user roles: administrators and students.

Administrators can upload student data via CSV files or manual input, generate PDF marksheets and certificates dynamically, and publish credentials to the blockchain.

Students receive automated email notifications with links to their documents and a verification page.

Any external verifier - an employer, government agency, or academic institution - can access the public verification interface without requiring authentication.

This report presents a comprehensive study of the Authblock system, detailing its objectives, scope, system design, algorithms, implementation, and results.

The system represents a significant step forward in the digitization and democratization of academic credential verification.

Chapter 2

OBJECTIVES OF THE PROJECT

The primary objectives of the Authblock mini-project are as follows:

To design and develop a decentralized application (dApp) that eliminates academic credential fraud by anchoring document hashes on the Ethereum blockchain, ensuring tamper-proof provenance.

To automate the generation of PDF marksheets and certificates from administrator-uploaded CSV data or manual entries, reducing manual effort and human error in cre

FAQs

What is GPTZero?

GPTZero is the leading AI detector for checking whether a document was written by a large language model such as ChatGPT. GPTZero detects AI on sentence, paragraph, and document level. Our model was trained on a large, diverse corpus of human-written and AI-generated text with support for English, Spanish, French, German, and other languages. To date, GPTZero has served over 17 million users around the world, and works with over 100 organizations in education, hiring, publishing, legal, and more.

When should I use GPTZero?

Our users have seen the use of AI-generated text proliferate into education, certification, hiring and recruitment, social writing platforms, disinformation, and beyond. We've created GPTZero as a tool to highlight the possible use of AI in writing text. In particular, we focus on classifying AI use in prose. Overall, our classifier is intended to be used to flag situations in which a conversation can be started (for example, between educators and students) to drive further inquiry and spread awareness of the risks of using AI in written work.

Does GPTZero only detect ChatGPT outputs?

No, GPTZero works robustly across a range of AI language models, including but not limited to ChatGPT, GPT-5, GPT-4, GPT-3, Gemini, Claude, and AI services based on those models.

What are the limitations of the classifier?

The nature of AI-generated content is changing constantly. As such, these results should not be used to punish students. We recommend educators to use our behind-the-scenes [Writing Reports](#) as part of a holistic assessment of student work. There always exist edge cases with both instances where AI is classified as human, and human is classified as AI. Instead, we recommend educators take approaches that give students the opportunity to demonstrate their understanding in a controlled environment and craft assignments that cannot be solved with AI. Our classifier is not trained to identify AI-generated text after it has been heavily modified after generation (although we estimate this is a minority of the uses for AI-generation at the moment). Currently, our classifier can sometimes flag other machine-generated or highly procedural text as AI-generated, and as such, should be used on more descriptive portions of text.

I'm an educator who has found AI-generated text by my students. What do I do?

Firstly, at GPTZero, we don't believe that any AI detector is perfect. There always exist edge cases with both instances where AI is classified as human, and human is classified as AI. Nonetheless, we recommend that educators can do the following when they get a positive detection: Ask students to demonstrate their understanding in a controlled environment, whether that is through an in-person assessment, or through an editor that can track their edit history (for instance, using our [Writing Reports](#) through Google Docs). Check out our list of [several recommendations](#) on types of assignments that are difficult to solve with AI.

Ask the student if they can produce artifacts of their writing process, whether it is drafts, revision histories, or brainstorming notes. For example, if the editor they used to write the text has an edit history (such as Google Docs), and it was typed out with several edits over a reasonable period of time, it is likely the student work is authentic. You can use GPTZero's Writing Reports to replay the student's writing process, and view signals that indicate the authenticity of the work.

See if there is a history of AI-generated text in the student's work. We recommend looking for a long-term pattern of AI use, as opposed to a single instance, in order to determine whether the student is using AI.